

PLANE LANDING ASISTANT USING FUZZY SYSTEM LOGIC

Student: Raul-George Horgos

Teacher: Florina Luminita Rostas

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INTRODUCTION

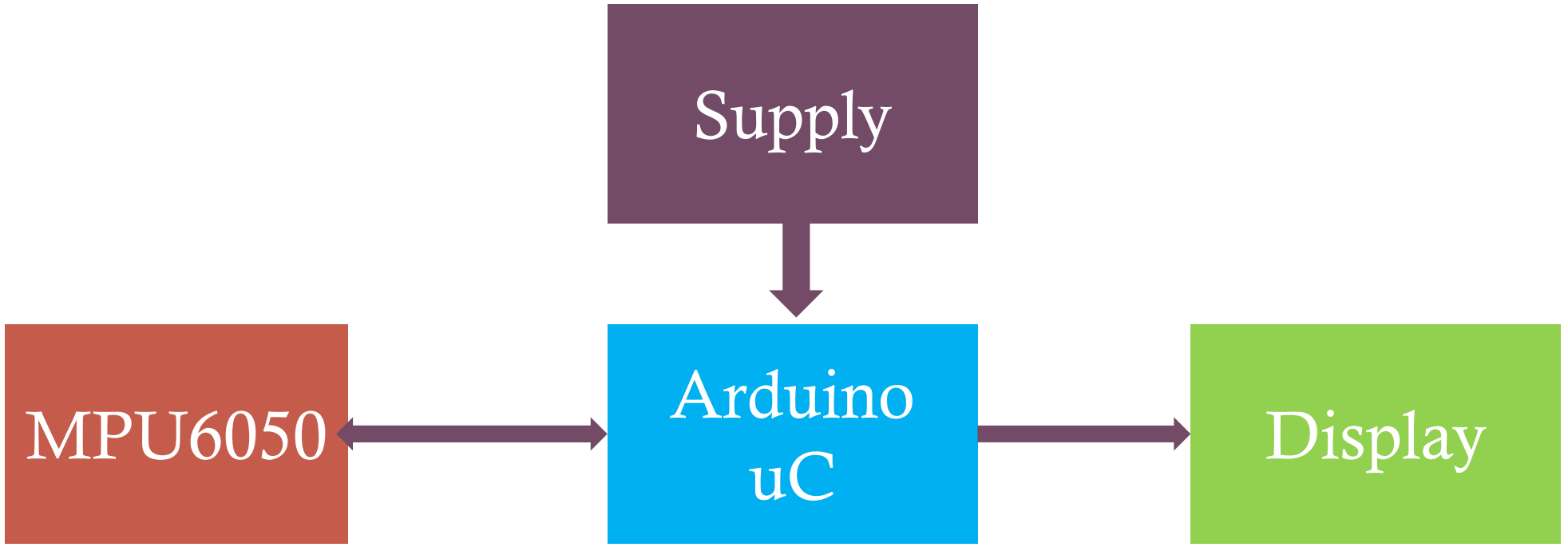
- This project develops a fuzzy logic-based landing assistant using an MPU6050 sensor, Arduino, and an LCD display. The MPU6050 provides real-time orientation data, which the fuzzy system processes to handle uncertain landing conditions and assist in stabilizing the plane during landing. The LCD displays information for angle adjutment.



CONTROL SYSTEM STRUCTURE

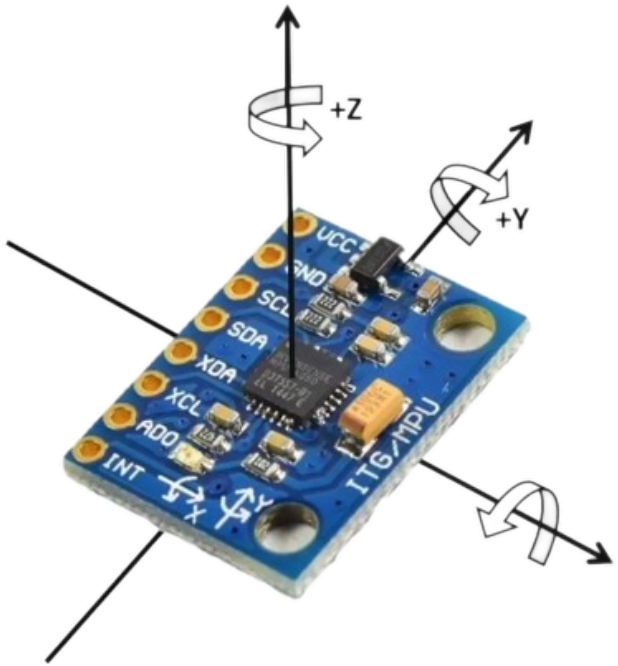
- Fuzzy controller using 9 rules
- MPU 6050 gyroscope and accelerometer
- Arduino Uno board
- LCD display using I2C protocol

BLOCK DIAGRAM



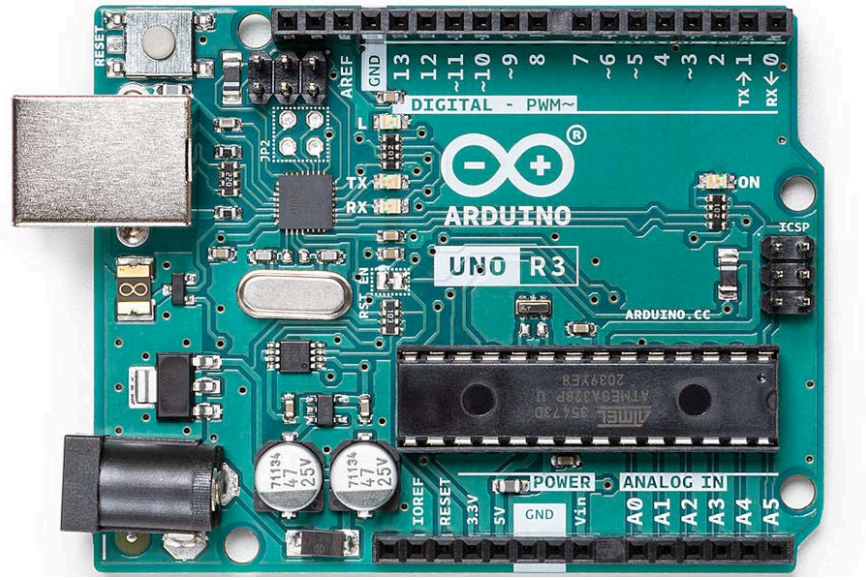
COMPONENTS
THE MPU6050

- The MPU6050 IMU has both 3-Axis accelerometer and 3-Axis gyroscope integrated on a single chip. The gyroscope measures rotational velocity or rate of change of the angular position over time, along the X, Y and Z axis.



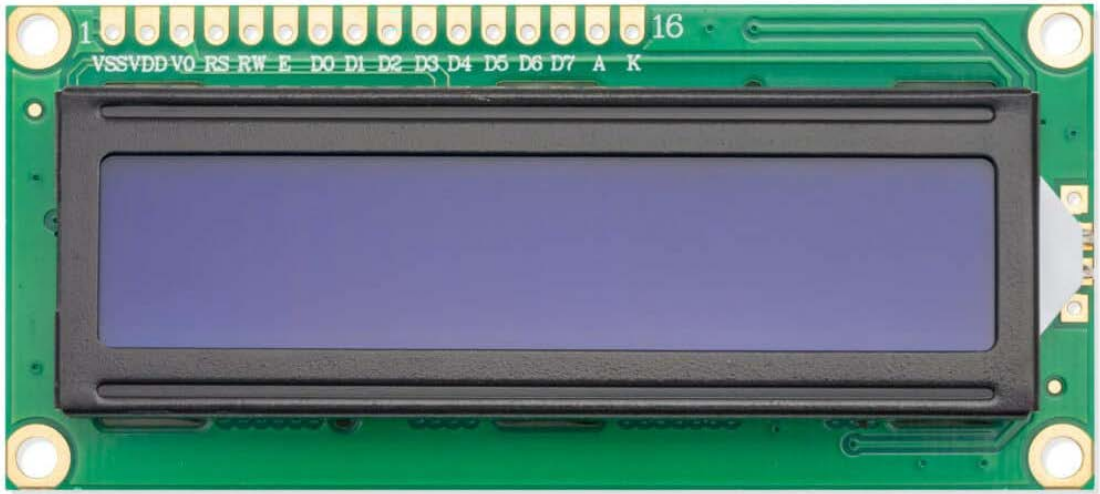
COMPONENTS
THE ARDUINO

- The Arduino Uno development board has a microcontroller board based on the ATmega328P. It has 14 digital input/output pins, 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button.

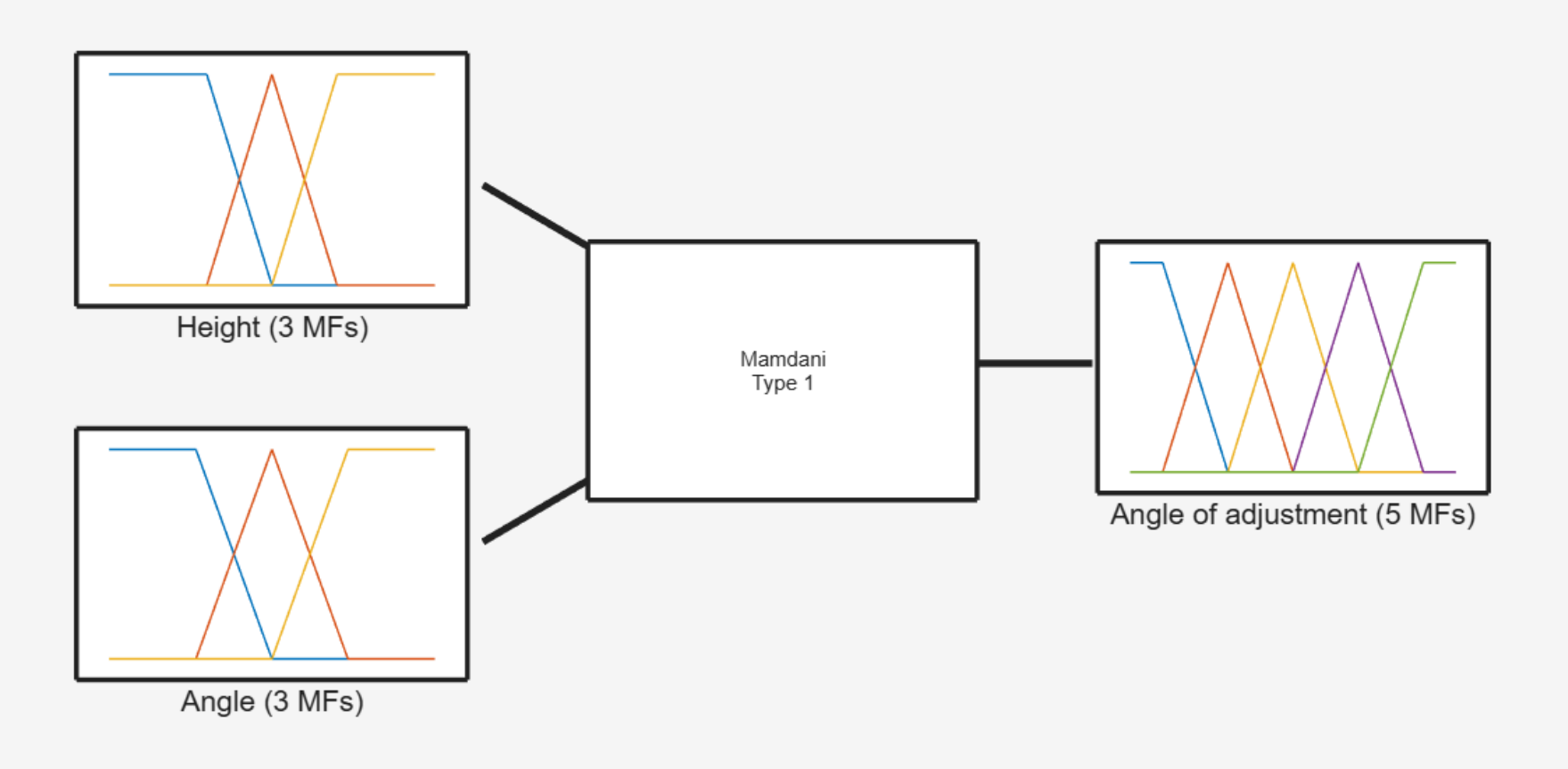


COMPONENTS
THE LCD DISPLAY

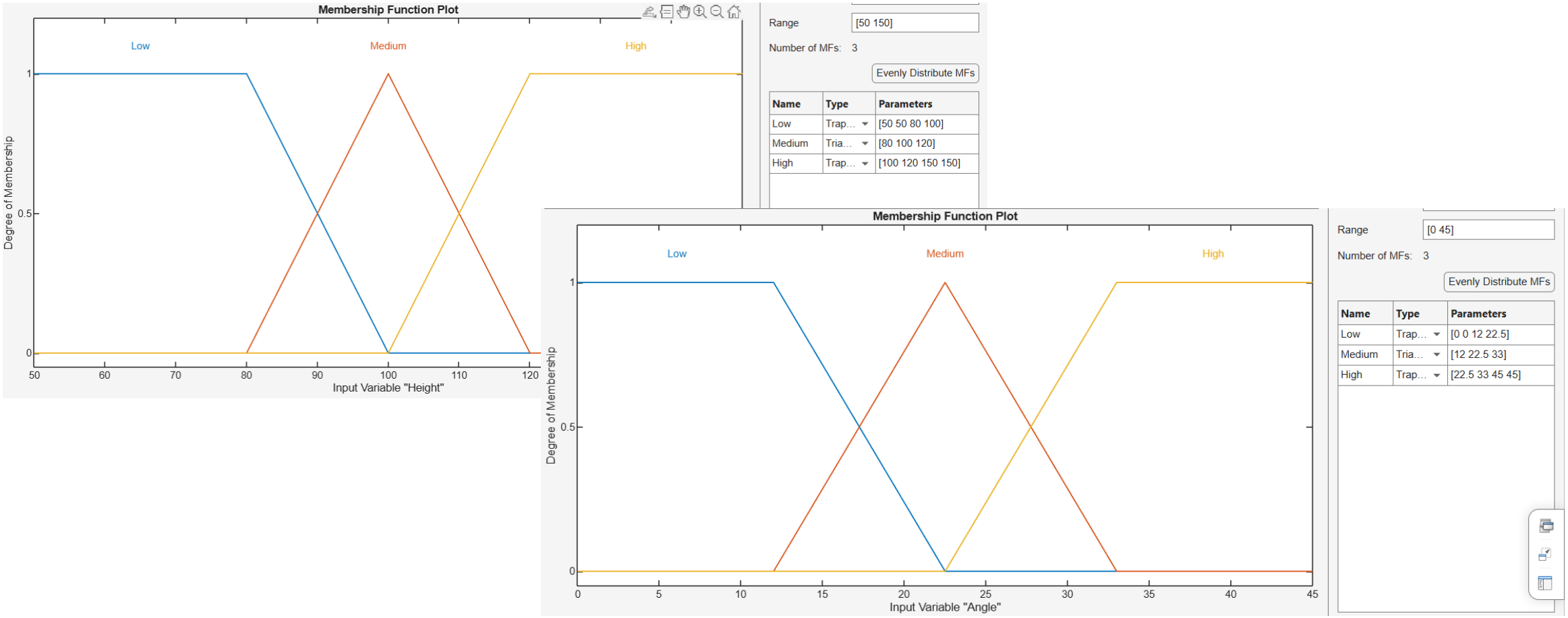
- The LCD provides real-time feedback on key flight parameters, for easy monitoring during landing and displays the percentage of the angle needed to be adjusted.



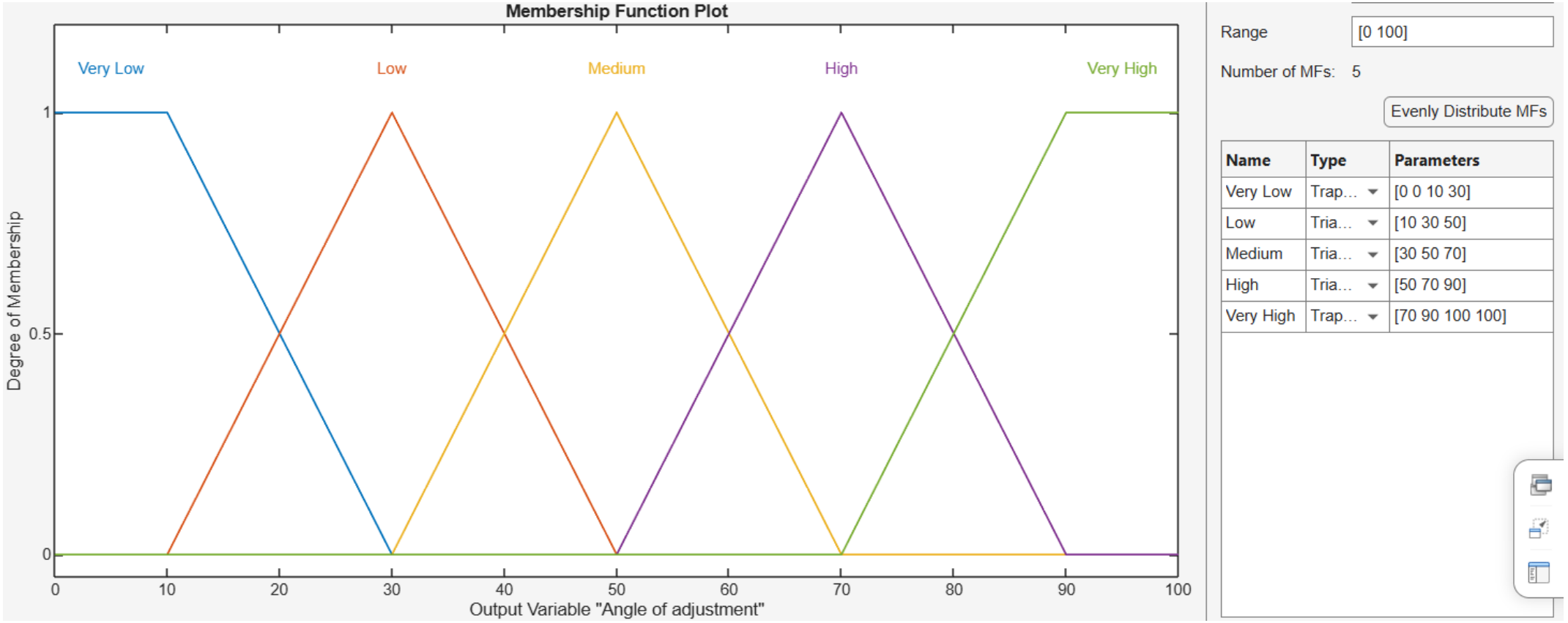
FUZZY LOGIC SYSTEM



FUZZY LOGIC SYSTEM: INPUTS



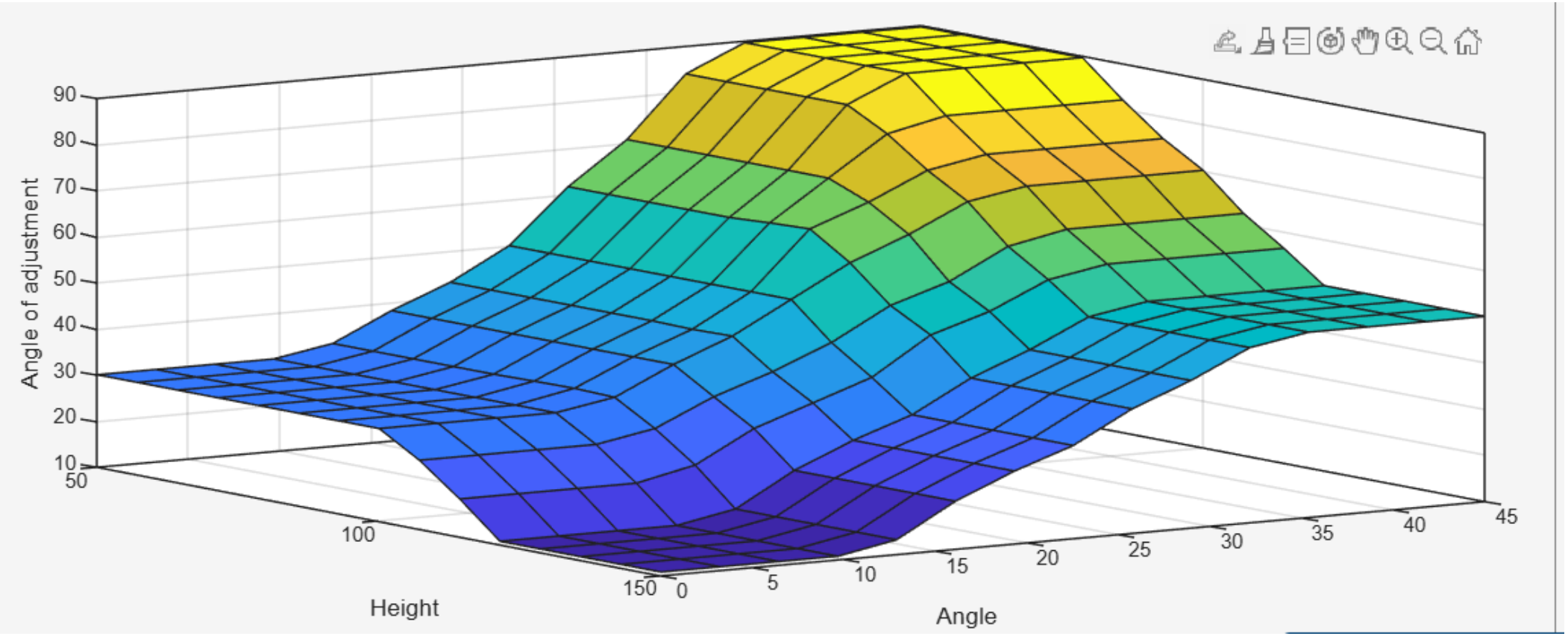
FUZZY LOGIC SYSTEM: OUTPUT



FUZZY LOGIC SYSTEM

Height / Angle	Low	Medium	High
Small	Small	Small	Very Small
Medium	Medium	Medium	Small
Big	Very Big	Big	Medium

FUZZY LOGIC SYSTEM



EXPERIMENTAL SETUP & RESULTS

PROPERTY EDITOR: INPUT

Name: Angle

Range: [0 45]

Number of MFs: 3

Evenly Distribute MFs

Name	Type	Parameters
Low	Trap...	[0 0 12 22.5]
Medium	Tria...	[12 22.5 33]
High	Trap...	[22.5 33 45 45]

```
float angleMFs[3][4] = {
    {0, 0, 12, 22.5},
    {12, 22.5, 33},
    {22.5, 33, 45, 45}
};
int angleMFTypes[3] = {1, 0, 1};
```

Name: Height

Range: [50 150]

Number of MFs: 3

Evenly Distribute MFs

Name	Type	Parameters
Low	Trap...	[50 50 80 100]
Medium	Tria...	[80 100 120]
High	Trap...	[100 120 150 150]

```
// Define membership functions
float heightMFs[3][4] = {
    {50, 50, 80, 100},
    {80, 100, 120},
    {100, 120, 150, 150}
};
int heightMFTypes[3] = {1, 0, 1};
```

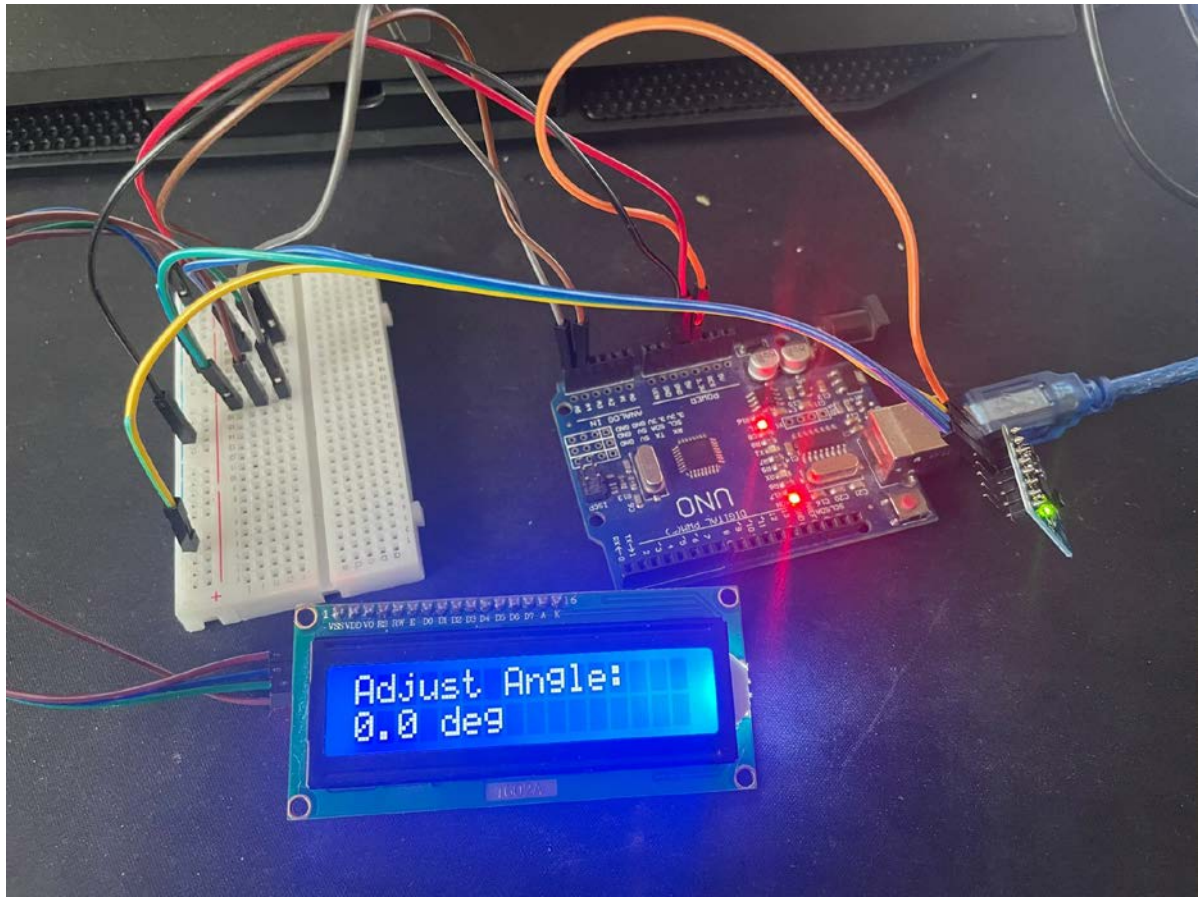
EXPERIMENTAL SETUP & RESULTS

	Rule	Weight	Name
1	If Height is Low and Angle is Low then Angle of adjustment is Low	1	rule1
2	If Height is Medium and Angle is Low then Angle of adjustment is Low	1	rule2
3	If Height is High and Angle is Low then Angle of adjustment is Very Low	1	rule3
4	If Height is Low and Angle is Medium then Angle of adjustment is Medium	1	rule4
5	If Height is Medium and Angle is Medium then Angle of adjustment is Medium	1	rule5
6	If Height is High and Angle is Medium then Angle of adjustment is Low	1	rule6
7	If Height is Low and Angle is High then Angle of adjustment is Very High	1	rule7
8	If Height is Medium and Angle is High then Angle of adjustment is High	1	rule8
9	If Height is High and Angle is High then Angle of adjustment is Medium	1	rule9

```
int rules[9][3] = {
    {0, 0, 1},
    {1, 0, 1},
    {2, 0, 0},
    {0, 1, 2},
    {1, 1, 2},
    {2, 1, 1},
    {0, 2, 4},
    {1, 2, 3},
    {2, 2, 2}
};
fuzzy.setRules(rules);
```



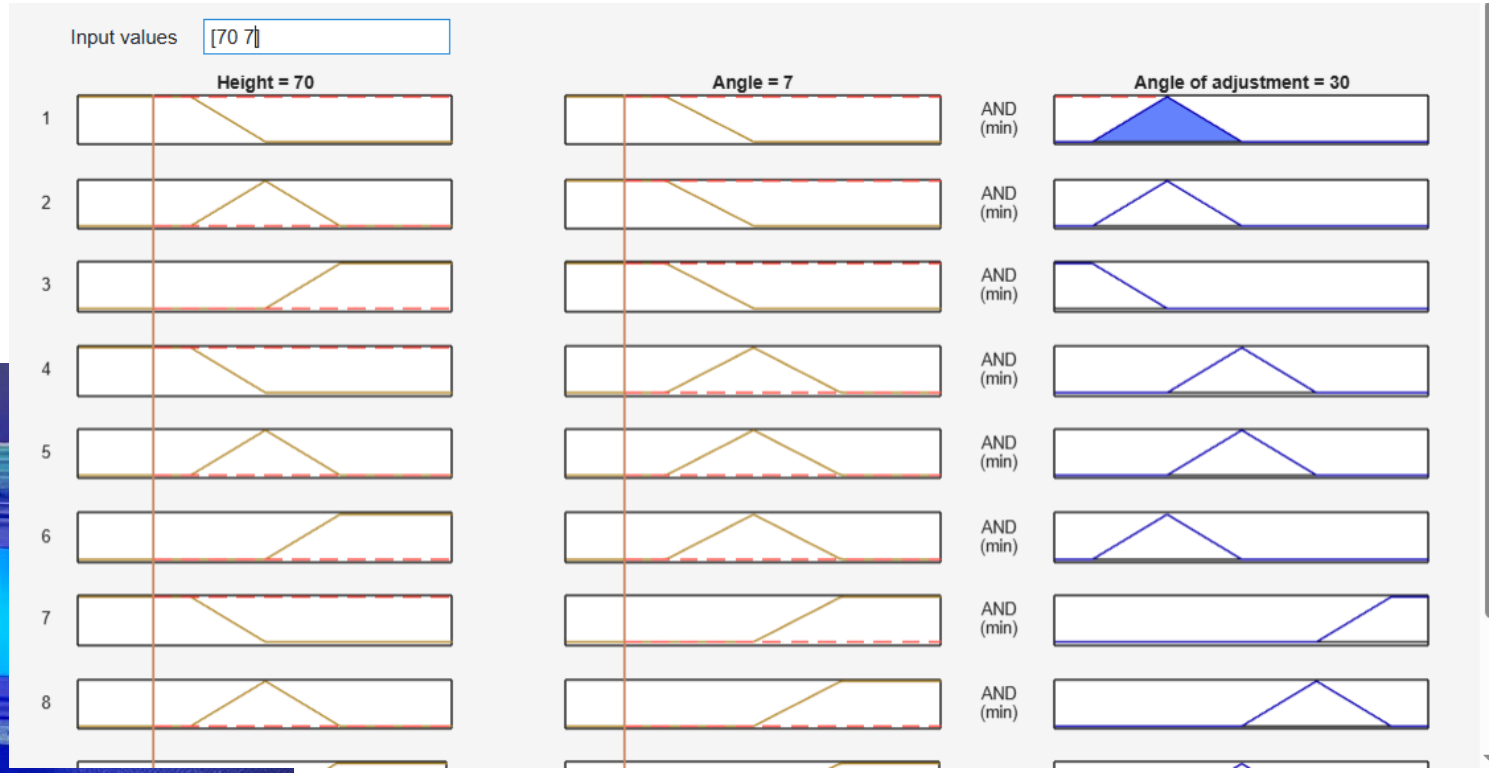
EXPERIMENTAL SETUP & RESULTS



EXPERIMENTAL SETUP & RESULTS

```
Height: 66.89 | Angle: 14.22
Height: 68.69 | Angle: 9.22
Height: 70.72 | Angle: 6.95
Height: 71.01 | Angle: 7.16
```





CONCLUSION

- Future improvements are always possible.
- As a conclusion I created a robust plane landing system using the fuzzy logic and implemented it hardware.

THANK YOU

